Nile Flow: Filtering and Smoothing with a Dynamic Linear Model

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1 Introduction

This analysis uses a dynamic linear model to describe changes in the underlying level of annual Nile flow. We distinguish sequential state estimation through Kalman filtering, one-step-ahead prediction, and retrospective reconstruction through smoothing.

We also examine sensitivity to the signal-to-noise ratio, check standardized forecast errors, and estimate the observation and state innovation variances by maximum likelihood.

2 Data and Model Specification

The Nile dataset, included in R, contains 100 annual observations of river flow at Aswan from 1871 to 1970, measured in $10^8\,\mathrm{m}^3$. The series shows a marked decline around the end of the nineteenth century.

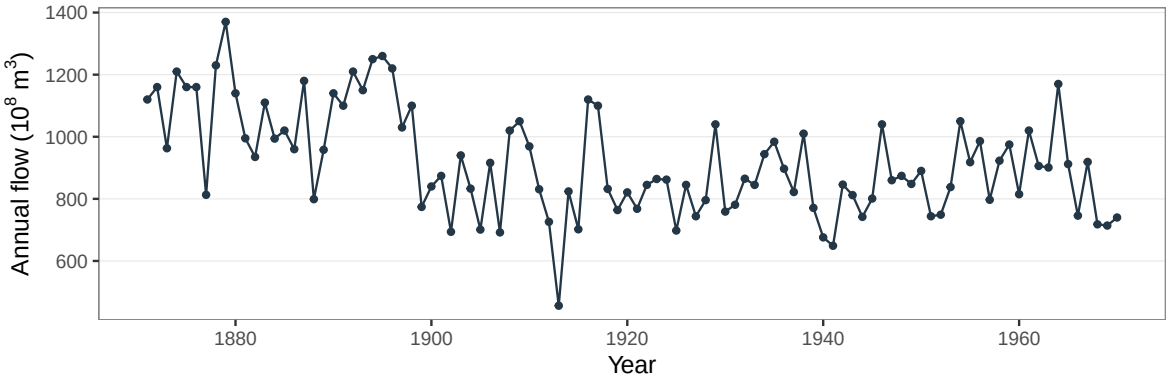


Figure 1. *Annual Nile flow at Aswan, 1871–1970. Data are available in R through the Nile dataset.*

We use a local level model, also called a random walk plus noise:

$$\begin{aligned} y_t &= \theta_t + v_t, \qquad v_t \sim \mathcal{N}(0, V), \\ \theta_t &= \theta_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, W). \end{aligned}$$

The observation errors and state innovations are independent over time and mutually independent, and are independent of the initial state.

For the initial analysis, the variances are treated as fixed at $V = 15100$ and $W = 1470$, with initial distribution

$$\theta_0 \sim \mathcal{N}(1000, 1000),$$

where the second parameter is the variance. The latent level can change over time, although the model does not explicitly specify a discrete change point.

3 Kalman Filtering

The Kalman filter updates the estimated level as each observation becomes available. The filtering distribution is

$$\theta_t \mid y_{1:t} \sim \mathcal{N}(m_t, C_t),$$

where m_t is the estimated level and C_t measures its remaining uncertainty after observing y_t.

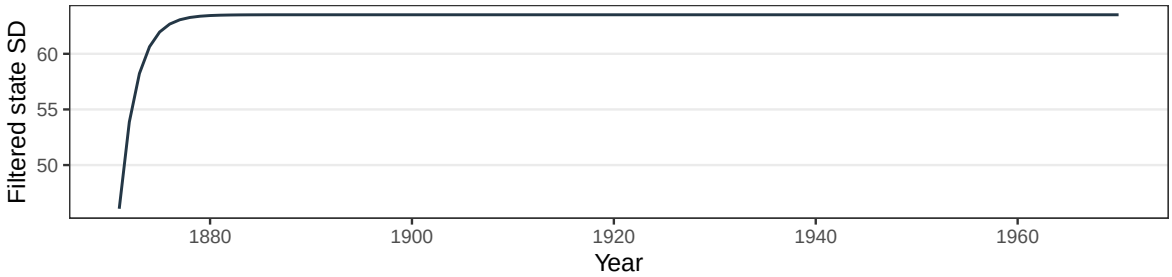


Figure 2. *Filtering standard deviation of the latent level, conditional on the specified variances and initial distribution.*

The filtering standard deviation initially increases and then stabilizes. The initial variance is below its steady-state value: new observations reduce uncertainty, but state innovations introduce additional uncertainty at every step. Consequently, the filtering variance does not converge to zero.

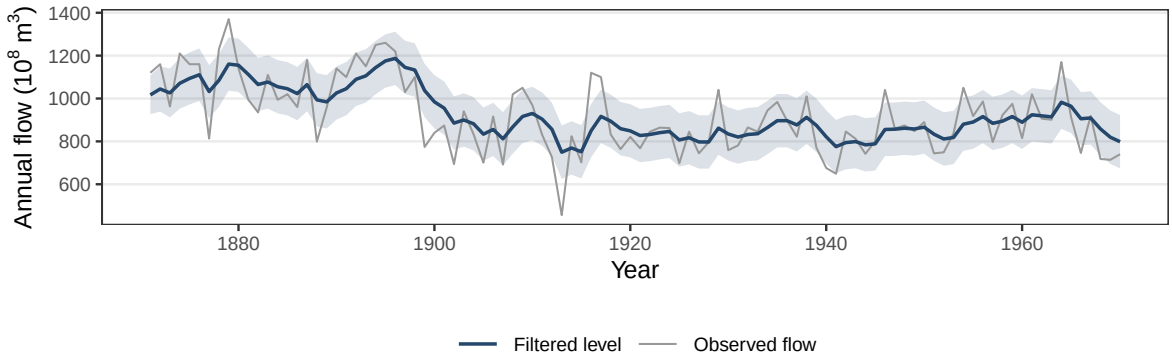


Figure 3. *Observed Nile flow and filtered latent level. The shaded band gives pointwise 95 percent credible intervals for the state, conditional on the specified model parameters.*

The filtered level follows the decline around the end of the nineteenth century while attenuating annual fluctuations. Because each estimate uses only observations available up to that year, adjustment to the decline occurs progressively.

The credible band becomes approximately constant in width as the filtering variance stabilizes. It describes uncertainty about the latent level, rather than the range of future observed flows; observations can therefore lie outside it without directly contradicting its stated coverage.

4 One-Step-Ahead Prediction

Before observing y_t, the model gives the predictive distribution

$$y_t \mid y_{1:t-1} \sim \mathcal{N}(f_t, Q_t), \qquad f_t = m_{t-1}, \qquad Q_t = C_{t-1} + W + V.$$

The forecast mean equals the previous filtered level. Predictive uncertainty includes uncertainty about that level, the next state innovation, and observation noise.

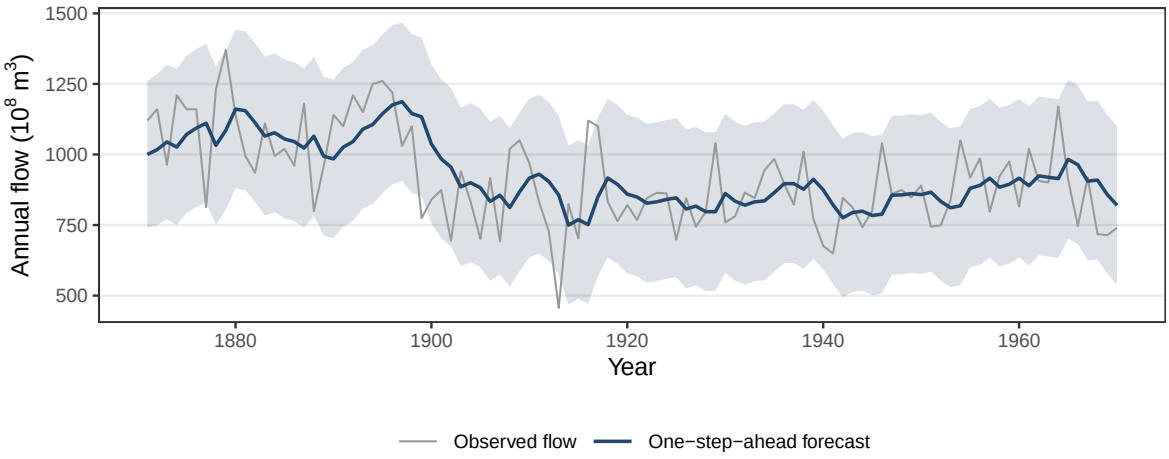


Figure 4. *Observed Nile flow and one-step-ahead forecast means. The shaded band gives pointwise 95 percent prediction intervals for observed flow, conditional on the specified model parameters.*

The forecasts follow the broad changes in flow but adjust with a delay, since each prediction equals the previous filtered level. This is particularly visible around the decline at the end of the nineteenth century.

Prediction intervals are wider than the filtering credible intervals because they describe uncertainty about an observation not yet seen and include observation noise. Their calibration will be examined through standardized forecast errors.

5 Signal-to-Noise Sensitivity

The ratio W/V influences how strongly the filter responds to new observations. We vary the observation variance V while holding W fixed, and then vary W while holding V fixed. The initial distribution remains unchanged.

The plot focuses on 1890–1950. These configurations illustrate sensitivity to the assumed variances; they are not selected through forecast performance.

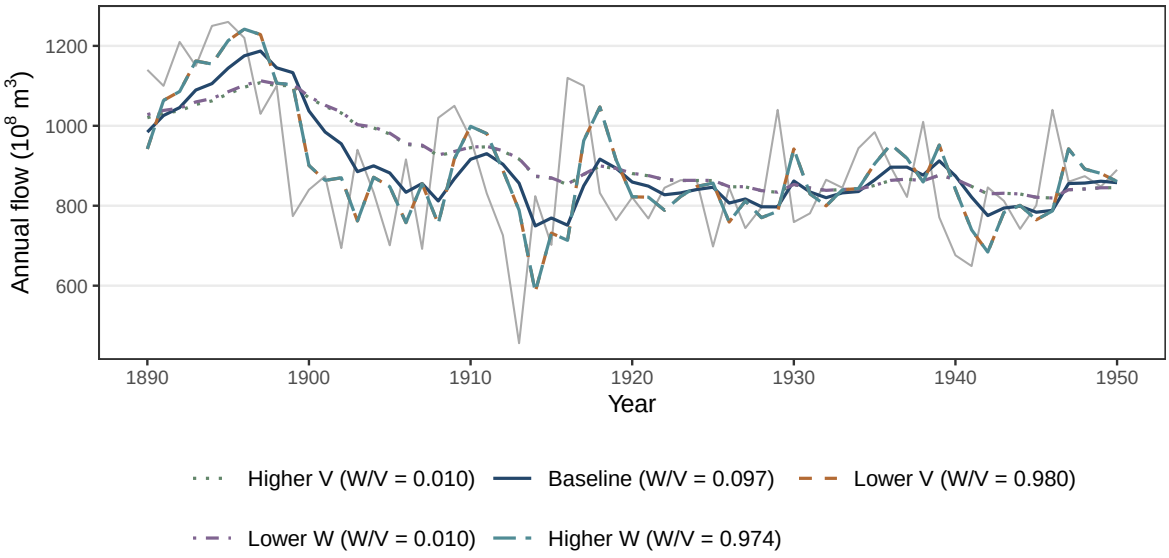


Figure 5. *One-step-ahead forecasts under five variance configurations, shown for 1890–1950. Observation variance V and state innovation variance W are varied separately around the baseline, keeping the initial distribution fixed.*

Increasing W or decreasing V gives new observations more weight in the filtering update, making subsequent forecasts more responsive to annual fluctuations. Decreasing W or increasing V produces smoother forecasts that adjust more slowly to changes in the observed level.

Similar ratios can produce similar forecast paths, so some curves may overlap. However, the ratio alone does not determine the uncertainty of the state estimates or forecasts, and the fixed initial variance also affects the early filtering updates.

Greater responsiveness does not itself imply better forecasts: this comparison illustrates the effect of the assumed variances, rather than establishing which configuration predicts best.

6 Forecast Error Diagnostics

We assess the baseline model using standardized one-step-ahead forecast errors:

$$z_t = \frac{y_t - f_t}{\sqrt{Q_t}}.$$

Under the specified Gaussian model with known parameters, these errors are independent standard normal variables. We examine their time pattern, autocorrelation, and normal quantile plot, together with a Ljung–Box test at lag 10.

These are model diagnostics on the observed sample, rather than an independent validation of the specified variances.

Table 1: Baseline forecast error diagnostics. Coverage is expressed as a proportion.

Diagnostic	Value
Mean standardized error	-0.044
SD of standardized errors	1.009
95 percent prediction interval coverage	0.950
Ljung-Box p-value (lag 10)	0.175

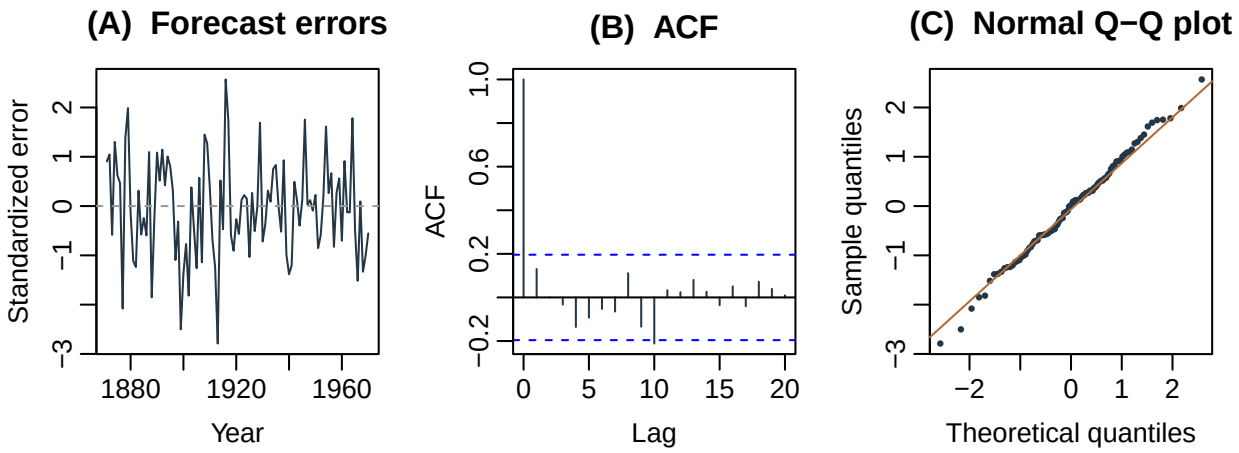


Figure 6. *Standardized forecast errors over time (A), their sample autocorrelation (B), and a normal quantile plot (C). Dashed ACF bounds are approximate pointwise 95 percent reference limits.*

The standardized errors have mean approximately -0.044 and standard deviation 1.009, close to the model reference values of zero and one. The prediction intervals contain 95 of the 100 observations.

The Ljung–Box test at lag 10 gives a p-value of 0.175, so it does not reject the absence of autocorrelation at the 5 percent level. The normal quantile plot is broadly linear, with some departures in the tails.

These checks reveal no strong overall contradiction of the baseline model, but do not establish independence, normality, or predictive calibration in other samples.

7 Smoothing

Smoothing estimates each past state using all observations, including those recorded after that year:

$$\theta_t \mid y_{1:T} \sim \mathcal{N}(s_t, S_t).$$

Unlike filtering, this is a retrospective reconstruction. The additional observations reduce state uncertainty: $S_t \leq C_t$. At the final observation, smoothing and filtering coincide because no subsequent data are available.

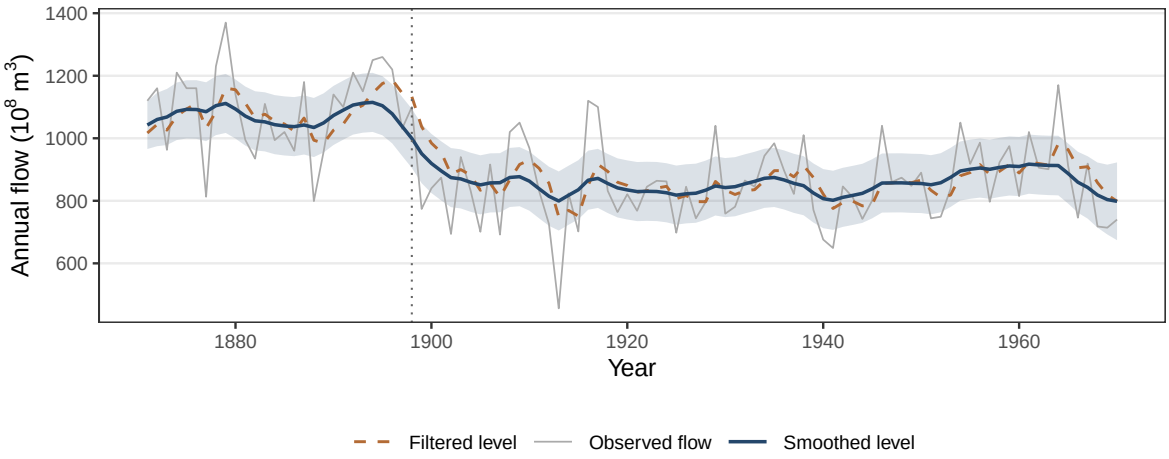


Figure 7. *Filtered and smoothed latent levels. The shaded band gives pointwise 95 percent smoothing credible intervals, conditional on the specified parameters. The vertical dotted line marks 1898 ($t = 28$).*

The smoothed level incorporates subsequent observations and therefore begins to decline earlier than the filtered level around 1898. This is a retrospective revision, rather than evidence that the model could anticipate the decline in real time.

The following comparison focuses on 1898 ($t = 28$), showing how subsequent observations revise the estimated latent level and its uncertainty.

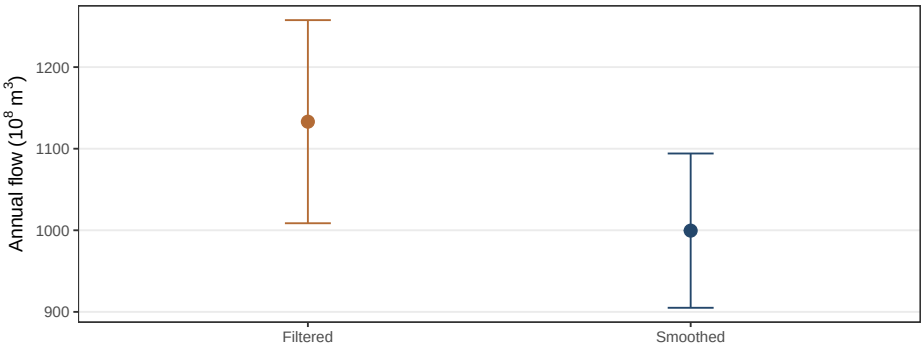


Figure 8. *Filtered and smoothed latent-level estimates for 1898, with pointwise 95 percent credible intervals conditional on the specified parameters.*

For 1898, the filtered level is 1133.1, with a 95 percent credible interval of [1008.6, 1257.6]. Using the full series revises the estimate to 999.6, with a narrower interval of [905.0, 1094.1]. The subsequent low-flow observations pull the retrospective estimate downward and reduce uncertainty about the latent level.

8 Variance Estimation

We now estimate the observation variance V and state innovation variance W by maximum likelihood, keeping the initial distribution fixed at $\theta_0 \sim \mathcal{N}(1000, 1000)$.

Optimization is performed on the logarithms of the variances to ensure positivity. Approximate standard errors are obtained from the inverse numerical Hessian and transformed to the variance scale using the delta method.

Table 2: Variance estimates conditional on the specified initial distribution.

Parameter	Specified value	MLE	Approx. SE
Observation variance V	15100	15000.90	3187.79
State innovation variance W	1470	1598.31	1348.67

The estimated variances are close to the specified values used in the filtering and smoothing analysis. These estimates are conditional on the fixed initial distribution $\theta_0 \sim \mathcal{N}(1000, 1000)$; a different initialization can produce different maximum-likelihood estimates.

The approximate standard error of W is large relative to its estimate, indicating substantial uncertainty about the state innovation variance. A symmetric normal confidence interval would extend below zero, so the standard error should be interpreted cautiously.

The state intervals reported earlier condition on the specified variance values and do not incorporate uncertainty about these parameters.

9 Discussion and Limitations

The local level model separates observed Nile flow into a changing latent level and observation noise. Filtering updates the level sequentially, while smoothing uses the full series to revise past estimates and reduce state uncertainty. The comparison for 1898 illustrates how subsequent low-flow observations alter the retrospective reconstruction.

The signal-to-noise ratio controls responsiveness: larger values produce faster adjustments to new observations. Forecast-error diagnostics reveal no strong overall contradiction with the model, but do not establish its adequacy in other samples.

The analysis assumes Gaussian, mutually independent observation and state innovations with constant variances. It does not explicitly model abrupt structural changes or their causes. Variance estimates and their approximate standard errors depend on this specification and the fixed initial distribution. The reported state intervals condition on the specified variances and exclude parameter uncertainty.